

# National Weather Service Prototype Hazard Notification Ladder

Exercise Reference Guide

| Product                      | Purpose                                                                                                                                                                                                                | When It Is Used                                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Space Weather Outlook        | Provides early awareness of a potential space weather threat several days in advance. Intended to support long-range planning and increase situational awareness before confidence is sufficient for hazard messaging. | Days before a possible event when a credible threat exists but confidence in impacts, timing, or magnitude remains low.                                                                          |
| Space Weather Statement      | Communicates important information about developing or changing conditions that do not yet require a Watch or Warning. Also used to provide updates or announce that an event has ended.                               | Used for developing events, lower-level hazards (G1-G2), significant scientific updates, or all-clear notifications.                                                                             |
| Space Weather Watch          | Alerts users that hazardous space weather conditions are possible and encourages preparedness. Communicates forecast uncertainty, expected timing, and the probability of hazardous conditions.                        | Issued when confidence has increased sufficiently that significant (G3+) impacts are possible but are not yet imminent or certain.                                                               |
| Space Weather Warning        | Indicates that hazardous space weather is expected or imminent. Provides the best estimate of timing, confidence, duration, and expected impacts to support protective actions.                                        | Issued when confidence is high that warning-level (G3+_ conditions will occur or have become imminent.                                                                                           |
| Observed Space Weather Alert | Announces that predefined hazard thresholds have been observed in real time. Provides rapid confirmation that hazardous conditions have begun.                                                                         | Issued immediately after operational observation thresholds are exceeded (for example, shock arrival or G-level thresholds). Discussion will be had about combining Alerts with Warning Updates. |
| Space Weather Notice         | Communicates operational information that affects interpretation of forecasts or observations but is not itself a hazard notification.                                                                                 | Used for data outages, degraded observations, instrument anomalies, delayed analyses, or other operational considerations.                                                                       |
| Space Weather Summary        | Provides a concise recap of the event, observed conditions, impacts, and expected recovery after hazardous conditions have subsided.                                                                                   | Issued following significant events to conclude operations and summarize the event.                                                                                                              |

# Exercise Discussion Guide

## National Weather Service Space Weather Product Prototypes

The purpose of this discussion is to gather feedback on the proposed hazard communication products, their operational usefulness, and opportunities for improvement. There are no right or wrong answers.

### Questions for All Participants

1. What was your overall impression of the prototype product suite?
2. Was the progression from Outlook → Statement → Watch → Warning → Observed Alert → Summary intuitive and easy to follow?
3. Did the products clearly communicate how confidence and urgency changed throughout the event?
4. Which product provided the greatest value to you, and why?
5. Was there information you expected to see that was missing?

### Forecaster Discussion

1. Would these products improve your ability to communicate forecast uncertainty?
2. Did the format make it easier or harder to prepare products compared to current operations?
3. Which portions of the products would you most likely edit before issuance?
4. Do you feel the product ladder appropriately supports operational decision making throughout the lifecycle of an event?
5. Are there portions of the workflow that could be automated to reduce forecaster workload?

\*If adopted operationally, approximately 90% of each text product would be automatically generated by the Geomagnetic Forecast Console, allowing forecasters to focus primarily on scientific analysis, quality control, and communicating forecast confidence rather than manually formatting text.

### Partner Discussion

1. Did the products clearly communicate what was happening and when impacts were expected?
2. Were the probabilities and uncertainty information useful for your planning or decision-making?
3. At what point in the product sequence would your organization begin taking action?
4. Did the partner email briefing and social media graphics complement the official text products?
5. Is there additional information or decision support that would improve the usefulness of these products for your organization?

### Closing Discussion

- What aspect of the prototype should be prioritized for further development?
- What concerns or barriers would you have to adopting this communication approach operationally?
- If you could change one thing before operational implementation, what would it be?
- Would this product suite improve communication between SWPC and its partners? Why or why not?
- Any additional comments or recommendations?

# Timeline Table - Gannon Storm - 06-16 May, 2024

| Issuance time            | Product                           | Purpose / event stage                                  | Status                  |
|--------------------------|-----------------------------------|--------------------------------------------------------|-------------------------|
| 1500 UTC Mon May 06 2024 | Space Weather Outlook             | Broader-horizon forecast and early awareness           | Included                |
| 0630 UTC Wed May 08 2024 | Space Weather Statement           | Solar eruption under analysis                          | Included                |
| 1600 UTC Wed May 08 2024 | Space Weather Statement           | Geomagnetic activity possible                          | Included                |
| 0800 UTC Thu May 09 2024 | Space Weather Watch               | Geomagnetic storm conditions possible                  | Included                |
| 1300 UTC Thu May 09 2024 | Space Weather Notice              | Delayed coronagraph imagery                            | Included                |
| 1230 UTC Fri May 10 2024 | Space Weather Warning             | Geomagnetic storm conditions expected                  | Included                |
| 1647 UTC Fri May 10 2024 | Observed Space Weather Alert      | Sudden impulse following coronal mass ejection arrival | Included                |
| 1722 UTC Fri May 10 2024 | Observed Space Weather Alert      | First G3 threshold attainment                          | Included                |
| Time not specified       | Observed Space Weather Alert — G4 | First G4 threshold attainment                          | Placeholder—not created |
| 2058 UTC Fri May 10 2024 | Observed Space Weather Alert      | First G5 threshold attainment                          | Included                |
| 0030 UTC Sat May 11 2024 | Warning Update Number 1           | Ongoing major geomagnetic storm                        | Included                |
| 1230 UTC Sat May 11 2024 | Warning Update Number 2           | Ongoing and renewed storm risk                         | Included                |
| 1400 UTC Sat May 11 2024 | Space Weather Notice              | Coronagraph contamination during solar particle storm  | Included                |
| 1800 UTC Sat May 11 2024 | Space Weather Summary Number 1    | Mid-event synthesis                                    | Included                |
| 0030 UTC Sun May 12 2024 | Warning Update Number 3           | Required 12-hour reassessment                          | Placeholder—not created |
| 1230 UTC Sun May 12 2024 | Warning Update Number 4           | Required 12-hour reassessment                          | Placeholder—not created |
| 0000 UTC Mon May 13 2024 | Space Weather Statement           | Ramp-down; Space Weather Warning ended                 | Included                |
| 1200 UTC Tue May 14 2024 | Space Weather Statement           | Warning-level all-clear                                | Included                |
| 1600 UTC Thu May 16 2024 | Space Weather Summary Number 2    | Final event summary and closure                        | Included                |

Note: IDSS briefings would begin with the Watch and be updated daily or as needed, typically alongside Warning Updates. Four email briefings will be created for the exercise: an initial broad core-partner briefing with the Watch; an updated broad core-partner briefing and tailored briefing with the first Warning; and a final broad core-partner briefing accompanying the all-clear.

# Space Weather Outlook

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER OUTLOOK  
National Weather Service Space Weather Prediction Center Boulder CO  
1500 UTC Mon May 06 2024

...ACTIVE SPACE WEATHER CONDITIONS EXPECTED THROUGH  
0000 UTC MON MAY 13 2024...

.DAY ONE...1500 UTC Mon May 06 2024 through  
0000 UTC Tue May 07 2024

Solar activity is expected to remain elevated. L2-L3 solar flare activity is expected, with intermittent HF radio degradation on the sunlit side of Earth. Additional L3 or greater flare activity is possible from the large, complex active regions on the visible disk.

Periods of G1-G2 geomagnetic storming are possible. Potential effects include degraded HF communication at higher latitudes, increased GNSS uncertainty in disturbed ionospheric regions, increased low-Earth-orbit satellite drag, and expanded aurora visibility where darkness and local viewing conditions allow.

SP1 or greater solar particle storm conditions are possible if additional L3 or greater flares occur.

.DAYS TWO THROUGH THREE...0000 UTC Tue May 07 2024 through  
0000 UTC Thu May 09 2024

Solar flare activity remains the primary concern. L2 activity is likely, and L3 activity remains possible. Geomagnetic activity is expected to decrease from Mon May 06 2024 levels, although unsettled to active periods may continue.

.DAYS FOUR THROUGH SEVEN...0000 UTC Thu May 09 2024 through  
0000 UTC Mon May 13 2024

Active Region 3664 remains large, complex, and capable of additional eruptions. Further Earth-directed coronal mass ejections are possible.

.PARTNER INFORMATION STATEMENT...

Partners should monitor SWPC updates through 0000 UTC Mon May 13 2024. A Space Weather Statement, Watch, or Warning may be issued if confidence increases for consequential geomagnetic, particle, or flare activity.

\$\$

## WHY THIS PROPOSAL HELPS

A structured Outlook preserves rapid weekly awareness while making the forecast drivers, time horizon, and escalation path clear.

- Explains the solar, geomagnetic, and particle environment rather than presenting scale values alone.
- Separates the near-term forecast from later-period uncertainty and identifies the potential for further eruptions.
- Creates an auditable pre-event record without inventing unsupported numerical probabilities.

# Space Weather Statement — eruption under analysis

Gannon storm reconstruction — representative prototype

## Proposed Product

WAXX22 KWNP 080630  
SWSXXX

SPACE WEATHER STATEMENT  
National Weather Service Space Weather Prediction Center Boulder CO  
0630 UTC Wed May 08 2024

...SOLAR ERUPTION UNDER ANALYSIS...

- \* WHAT...A significant solar flare occurred early this morning from a large, complex active region. Initial observations suggest the flare may have produced a coronal mass ejection (CME), which could result in geomagnetic storms if confirmed.
- \* WHERE...Worldwide and Earth's orbit.
- \* WHEN...Potential Earth impacts are possible later this week. Timing remains uncertain while imagery and model analysis continue.
- \* CONFIDENCE...Low. Early imagery and the location of the eruption suggest an Earth-directed trajectory is possible.
- \* IMPACTS...If an Earth-directed coronal mass ejection is confirmed, geomagnetic activity could affect HF communications, satellite navigation, satellite operations, and electric power systems, particularly at high latitudes.
- \* ADDITIONAL DETAILS...No Space Weather Watch or Warning is in effect for this event at this time. A Space Weather Watch may be issued if additional imagery and model guidance indicate possible significant impacts to Earth.
- \* NEXT UPDATE...No later than 1800 UTC Wed May 08 2024, or sooner if conditions change.

\$\$

## WHY THIS PROPOSAL HELPS

An early-analysis Statement gives partners an authoritative status while SWPC determines whether an eruption is Earth-directed.

- Confirms that SWPC has identified the event and analysis is underway.
- States what is known, what remains unresolved, and when the next update will arrive.
- Establishes a clear path to a Watch without prematurely requesting operational action.

Special Weather Statements heighten awareness of ongoing, imminent, or developing conditions. This proposed use applies that established NWS function to the early, still-uncertain phase of a space weather event. [NWSI 10-517](#)

# Space Weather Statement — G1-G2 conditions

Gannon storm reconstruction — representative prototype

## Proposed Product

WAXX22 KWNP 081600  
SWSXXX

SPACE WEATHER STATEMENT  
National Weather Service Space Weather Prediction Center Boulder CO  
1600 UTC Wed May 08 2024

...GEOMAGNETIC ACTIVITY POSSIBLE FROM 1200 UTC FRI MAY 10 2024 THROUGH  
1200 UTC MON MAY 13 2024...

- \* WHAT...Geomagnetic storm (G1-G2) likely.
- \* WHERE...Worldwide and Earth's orbit. Higher-latitude regions are most likely to experience operational effects.
- \* WHEN...There is a 50% chance that G1 or greater conditions will begin between 1800 UTC Fri May 10 2024 and 0600 UTC Sat May 11 2024.  
  
Earliest Storm Onset: 1200 UTC Fri May 10 2024  
Latest Storm Onset: 1200 UTC Sun May 12 2024  
  
Minimum Duration: 12 hours  
Maximum Duration: 24 hours
- \* CONFIDENCE...There is a 60% chance that the peak will reach G2 and a 10% chance that it will reach G3 or greater.
- \* IMPACTS...Power grids may experience small voltage changes or alarms. Increased drag may reduce the accuracy of low-Earth-orbit satellite predictions. Long-distance HF communication may fade at higher latitudes. Aurora may expand across more of the northern United States where darkness, clouds, and local viewing conditions allow.
- \* ADDITIONAL DETAILS...Solar activity remains high. The forecast for elevated geomagnetic activity is based on two coronal mass ejections, but additional eruptions are possible. No Space Weather Watch or Warning is in effect at 1600 UTC Wed May 08 2024. A Watch may be issued if confidence increases for G3 or greater conditions. Partners should continue to monitor SWPC products for updated information.
- \* NEXT UPDATE...No later than 1000 UTC Thu May 09 2024, or sooner if conditions change.

\$\$

## WHY THIS PROPOSAL HELPS

A G1-G2 Statement preserves early awareness while keeping the action level proportional to the anticipated hazard.

- Provides an explicit onset window, plausible bounds, and duration.

- Separates onset probability from conditional storm-intensity probabilities.
- Explains impacts, escalation criteria, and the next decision point.

This Statement carries lower-level hazard information below Watch and Warning criteria while preserving a defined escalation path. [NWSI 10-517](#)

Ensemble guidance

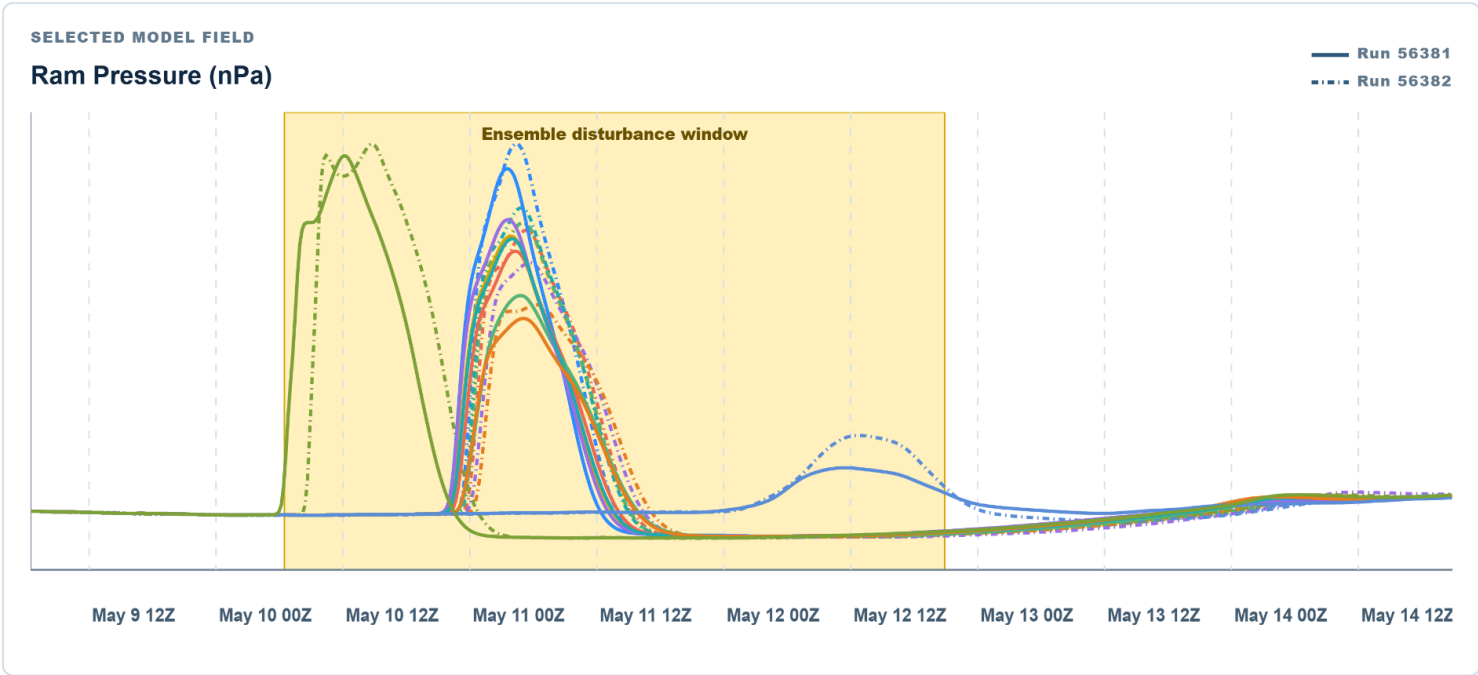
Runs 56381 and 56382 have fit/median arrivals near 2244 UTC Fri May 10 2024 and 2348 UTC Fri May 10 2024. Their centers are about one hour apart, increasing confidence that the primary onset window is from 1800 UTC Fri May 10 2024 through 0600 UTC Sat May 11 2024. Closely spaced coronal mass ejection effects are possible. The public probability is a forecaster assessment informed by the ensemble, not a raw member fraction.

NOAA SPACE WEATHER PREDICTION CENTER

WSA-Enlil Ensemble Guidance



|                                                                                                             |                                                                                            |                                                                        |                                                        |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|
| SCENARIO ARRIVAL WINDOW<br><b>0627 UTC Fri May 10 2024</b><br>through 0533 UTC Sun May 12 2024 latest onset | SCENARIO MEDIAN ARRIVAL<br><b>2333 UTC Fri May 10 2024</b><br>Middle scenario arrival time | TIMING SPREAD<br><b>47.1 hr</b><br>Earliest to latest scenario arrival | PARAMETER RANGE<br><b>0.0–15.1 nPa</b><br>Ram Pressure |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|



# Space Weather Watch — G3-G4 conditions possible

Gannon storm reconstruction — representative prototype

## Proposed Product

WOXX22 KWNP 090800  
WATA50

URGENT - SPACE WEATHER MESSAGE  
SPACE WEATHER WATCH  
National Weather Service Space Weather Prediction Center Boulder CO  
0800 UTC Thu May 09 2024

...SPACE WEATHER WATCH IN EFFECT FOR GEOMAGNETIC STORM CONDITIONS FROM  
1200 UTC FRI MAY 10 2024 THROUGH 0600 UTC MON MAY 13 2024...

- \* WHAT...Geomagnetic storm (G3-G4) possible.
- \* WHERE...Worldwide and Earth's orbit. Higher-latitude regions face the greatest potential impacts.
- \* WHEN...There is a 70% chance that storm onset will occur between 1800 UTC Fri May 10 2024 and 0600 UTC Sat May 11 2024.

Earliest Storm Onset: 1200 UTC Fri May 10 2024  
Latest Storm Onset: 0600 UTC Sat May 11 2024

Minimum Duration: 12 hours  
Maximum Duration: 48 hours

- \* CONFIDENCE...There is a 65% chance of G3 or greater conditions and a 20% chance of G4 or greater conditions.
- \* IMPACTS...Power systems may experience voltage-control issues, transformer stress, and isolated equipment trips. Atmospheric drag may lower satellite orbits and reduce the accuracy of tracking and orbit predictions. HF communication, precision satellite navigation, and some satellite communications may be intermittently degraded, particularly at higher latitudes. Geomagnetically induced currents may affect pipeline monitoring and corrosion-control systems. Aurora may be visible across much of the United States where darkness and local viewing conditions allow.

### PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should review sector-specific procedures, monitor SWPC updates, and consult relevant decision-support guidance.

- \* ADDITIONAL DETAILS...Additional ensemble members indicate possible coronal mass ejection influences beyond the primary onset window. A Warning may be issued if G3 or greater conditions become expected.
- \* NEXT UPDATE...No later than 2000 UTC Thu May 09 2024, or sooner if conditions change.

\$\$

WHY THIS PROPOSAL HELPS

A Watch reserved for possible G3 or greater conditions gives partners a clear preparation signal before warning criteria are met.

- Pairs the most-likely onset window with earliest and latest plausible timing.
- Uses monotonic G3-or-greater and G4-or-greater probabilities.
- Translates the forecast into preparation actions and sector-specific consequences.

Ensemble guidance

The ensemble signal has strengthened from one or two possible arrivals into a multi-disturbance scenario. Runs 56381, 56382, and 56385 indicate Earth-arriving disturbances, with primary fit/median arrivals from 2244 UTC Fri May 10 2024 through 0623 UTC Sat May 11 2024.

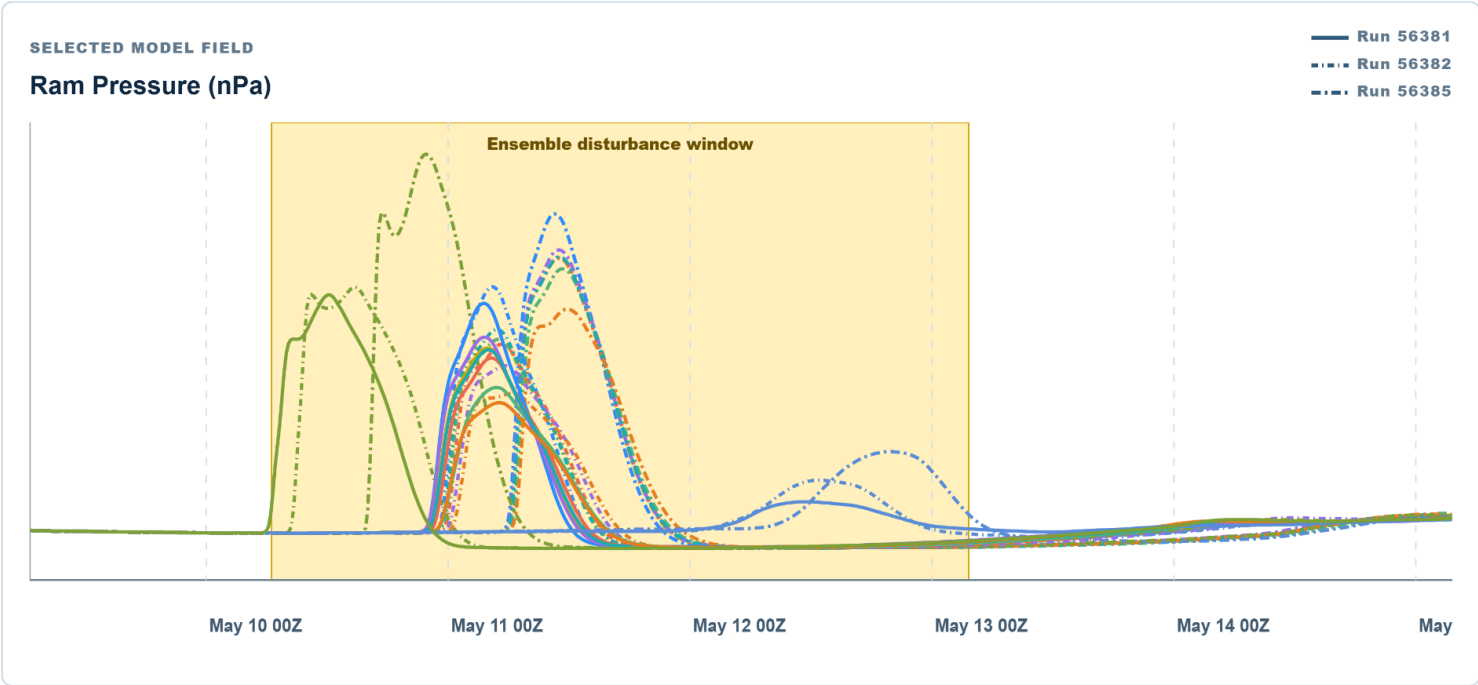
Runs 56381 and 56382 are tightly grouped near 2244 UTC Fri May 10 2024 and 2348 UTC Fri May 10 2024. Run 56385 centers near 0623 UTC Sat May 11 2024. The wider ensemble spread extends from 1200 UTC Fri May 10 2024 into Sun May 12 2024, supporting a focused most-likely onset window while retaining broader uncertainty in subsequent influences.

NOAA SPACE WEATHER PREDICTION CENTER

WSA-Enlil Ensemble Guidance



|                                                                                                             |                                                                                            |                                                                        |                                                        |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|
| SCENARIO ARRIVAL WINDOW<br><b>0627 UTC Fri May 10 2024</b><br>through 1051 UTC Sun May 12 2024 latest onset | SCENARIO MEDIAN ARRIVAL<br><b>2347 UTC Fri May 10 2024</b><br>Middle scenario arrival time | TIMING SPREAD<br><b>52.4 hr</b><br>Earliest to latest scenario arrival | PARAMETER RANGE<br><b>0.0–22.7 nPa</b><br>Ram Pressure |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|



# Space Weather Notice — delayed coronagraph imagery

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER NOTICE  
National Weather Service Space Weather Prediction Center Boulder CO  
1300 UTC Thu May 09 2024

...CORONAGRAPH IMAGERY AVAILABILITY AFFECTING CORONAL MASS  
EJECTION ANALYSIS...

\* WHAT...Coronagraph imagery needed for additional coronal mass ejection  
analysis has not updated since 0905 UTC Thu. SWPC is continuing to analyze  
recent eruptions using the imagery and guidance available at  
1300 UTC Thu May 09 2024.

\* IMPACTS...Confidence in the exact arrival timing, duration, and strength  
of potential coronal mass ejection impacts may be reduced until  
additional imagery becomes available. The overall forecast continues  
to support elevated to significant geomagnetic activity from  
1200 UTC Fri May 10 2024 into Sat May 11 2024.

\* ADDITIONAL DETAILS...The active Space Weather Watch remains in effect.  
This Notice describes a forecast-data limitation and does not change  
the hazard assessment.

\* NEXT UPDATE...The imagery limitation will be addressed in the next  
Space Weather Watch update no later than  
2000 UTC Thu May 09 2024, or sooner if availability changes.

\$\$

## WHY THIS PROPOSAL HELPS

A Space Weather Notice makes forecast-data limitations visible without changing the active hazard product.

- Explains exactly which observation is delayed and which forecast elements may be less precise.
- Prevents users from interpreting reduced data quality as a reduction in hazard.
- Defines when and where SWPC will provide the next status update.

# Space Weather Warning — G3-G5

Gannon storm reconstruction — representative prototype

## Proposed Product

WOXX22 KWNP 101230  
WATA50

URGENT - SPACE WEATHER MESSAGE  
SPACE WEATHER WARNING  
National Weather Service Space Weather Prediction Center Boulder CO  
1230 UTC Fri May 10 2024

...SPACE WEATHER WARNING IN EFFECT FOR GEOMAGNETIC STORM CONDITIONS FROM  
1200 UTC FRI MAY 10 2024 THROUGH 0600 UTC MON MAY 13 2024...

- \* WHAT...Geomagnetic storm (G3-G4) expected.
- \* WHERE...Worldwide and Earth's orbit. Middle- and higher-latitude regions face the greatest potential effects.
- \* WHEN...There is a 65% chance that storm onset will occur between 1800 UTC Fri May 10 2024 and 0000 UTC Sat May 11 2024.

Earliest Storm Onset: 1230 UTC Fri May 10 2024  
Latest Storm Onset: 0600 UTC Sat May 11 2024

Minimum Duration: 24 hours  
Maximum Duration: 48 hours

- \* CONFIDENCE...There is an 85% chance of G3 or greater conditions, including a 5% chance of G5 conditions.
- \* IMPACTS...Power systems may experience voltage-control issues, transformer stress, and isolated equipment trips. Atmospheric drag may lower satellite orbits and reduce the accuracy of tracking and orbit predictions. HF communication, precision satellite navigation, and some satellite communications may be intermittently degraded, particularly at higher latitudes. Geomagnetically induced currents may affect pipeline monitoring and corrosion-control systems. Aurora may be visible across much of the United States and potentially farther south where darkness and local viewing conditions allow.

### PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should prepare for significant geomagnetic activity, follow sector-specific procedures, and maintain heightened situational awareness through 0030 UTC Sat May 11 2024.

- \* ADDITIONAL DETAILS...This Warning replaces the previous Space Weather Watch. Additional coronal mass ejection influences may produce renewed geomagnetic activity from 0000 UTC Sat May 11 2024 through 0600 UTC Mon May 13 2024.
- \* NEXT UPDATE...No later than 0030 UTC Sat May 11 2024, or sooner if conditions change.

\$\$

WHY THIS PROPOSAL HELPS

A single event-based Warning gives users time to act before G3-G5 impacts begin.

- Replaces a stream of technical K-index messages with one coherent hazard assessment.
- Distinguishes the probability of storm occurrence from the probability of the narrower onset window.
- Provides severity, duration, impacts, partner actions, and an exact update deadline.

Ensemble guidance supporting the Warning

Runs 56381 and 56382 show closely clustered fit/median arrivals near 2244 UTC Fri May 10 2024 and 2348 UTC Fri May 10 2024. Run 56385 adds an expected arrival near 0623 UTC Sat May 11 2024, and run 56387 extends the concern to near 0159 UTC Sun May 12 2024. Together, the runs support a multi-disturbance sequence rather than a single short-duration event.

The operational signal is that multiple disturbances are favorably positioned to reach Earth within a compressed period, with additional influence possible into Sun May 12 2024. That supports escalation to a Warning because the event has a credible path to G3-G4 conditions, with a lower-probability G5 outcome.

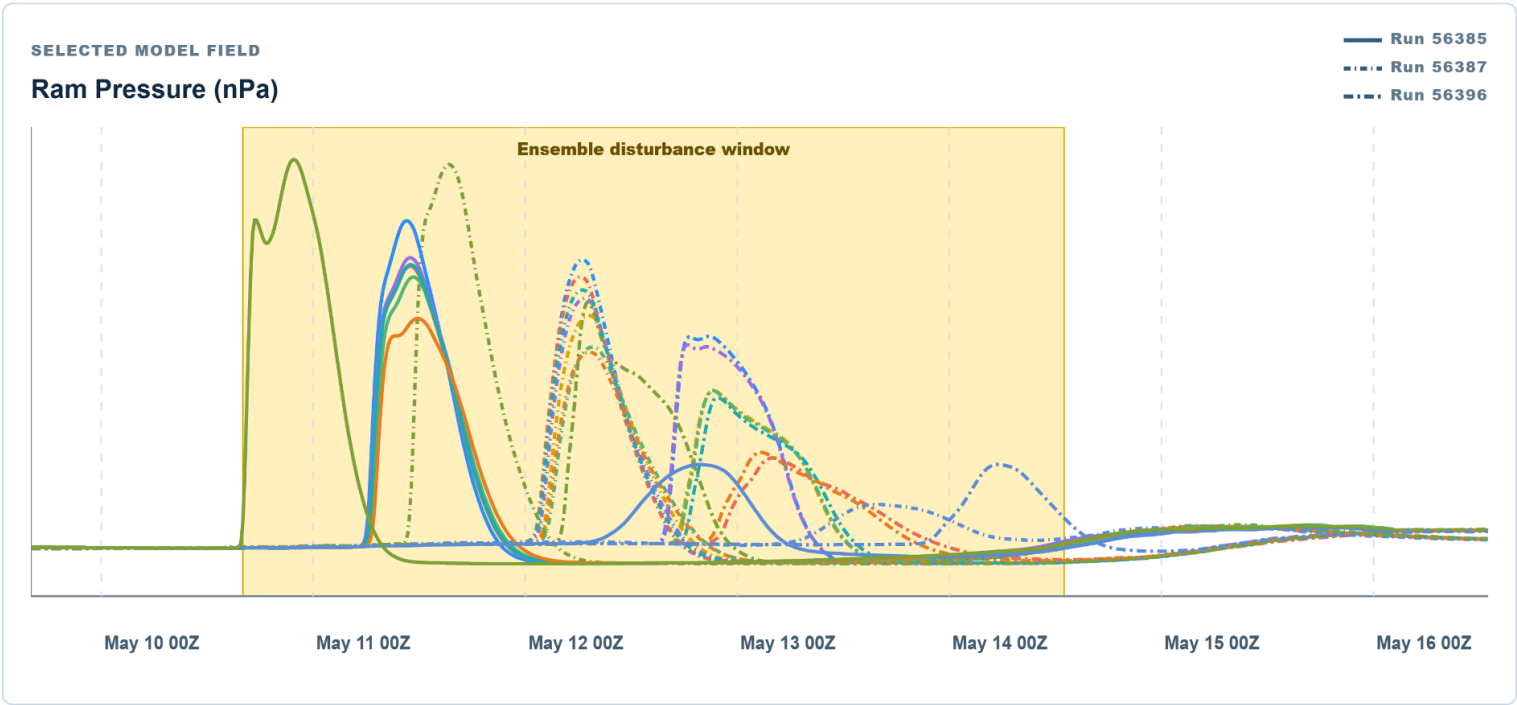
The ensemble supports the Warning's most-likely onset window while retaining earlier and later bounds. A Warning issued at 1230 UTC Fri May 10 2024 would provide about four hours of lead time before upstream shock detection at 1639 UTC Fri May 10 2024 and the sudden impulse at 1645 UTC Fri May 10 2024.

NOAA SPACE WEATHER PREDICTION CENTER

# WSA-Enlil Ensemble Guidance



|                                                                                                      |                                                                                     |                                                                 |                                                 |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------|
| SCENARIO ARRIVAL WINDOW<br>1601 UTC Fri May 10 2024<br>through 2327 UTC Mon May 13 2024 latest onset | SCENARIO MEDIAN ARRIVAL<br>0214 UTC Sun May 12 2024<br>Middle scenario arrival time | TIMING SPREAD<br>79.4 hr<br>Earliest to latest scenario arrival | PARAMETER RANGE<br>0.0–22.7 nPa<br>Ram Pressure |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------|



# Observed Space Weather Alert — shock arrival

Gannon storm reconstruction — representative prototype

## Proposed Product

OBSERVED SPACE WEATHER ALERT  
National Weather Service Space Weather Prediction Center Boulder CO  
1647 UTC Fri May 10 2024

...SUDDEN IMPULSE OBSERVED FOLLOWING CORONAL MASS EJECTION  
ARRIVAL...

\* WHAT...A geomagnetic sudden impulse has been observed.

\* OBSERVED...The sudden impulse was observed at 1645 UTC Fri May 10 2024 following detection of an interplanetary shock in the upstream solar wind at 1639 UTC Fri May 10 2024.

\* BASIS...This alert is based solely on real-time upstream solar wind measurements available at 1647 UTC Fri May 10 2024.

\* CURRENT STATUS...Solar wind conditions are becoming strongly enhanced following shock arrival. Geomagnetic activity is expected to increase as the disturbance moves past Earth. A Space Weather Warning remains in effect for G3-G5 conditions.

\* FUTURE ALERTS...This alert confirms arrival of a coronal mass ejection-driven disturbance. Additional Observed Space Weather Alerts will be issued when geomagnetic activity first reaches G3, G4, or G5.

\* IMPACTS...The sudden impulse may produce rapid magnetic-field changes and marks the beginning of potentially significant geomagnetic activity. Power grid, satellite, GNSS, HF radio, and other magnetically sensitive operations should prepare for worsening conditions.

PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should begin heightened monitoring, follow sector-specific procedures, and monitor SWPC Warning Updates, Observed Space Weather Alerts, and decision-support guidance.

\$\$

## WHY THIS PROPOSAL HELPS

An Observed Space Weather Alert correctly identifies the sudden impulse as confirmation of arrival rather than a forecast Warning.

- Connects upstream shock detection and the ground response to the active forecast.
- Signals the transition from preparation to heightened monitoring.
- Explains which first-threshold alerts will follow as the event escalates.

# Observed Space Weather Alert — G3

Gannon storm reconstruction — representative prototype

## Proposed Product

```
OBSERVED SPACE WEATHER ALERT
National Weather Service Space Weather Prediction Center Boulder CO
1720 UTC Fri May 10 2024

...GEOMAGNETIC STORM CONDITIONS OBSERVED...

* WHAT...Geomagnetic activity has reached the G3 (Strong) level.

* OBSERVED...G3 conditions were observed at 1720 UTC during the
  1500-1800 UTC observation period.

* BASIS...This alert is based on 12/13 real-time observations from the
  official Kp magnetometer network available at the time of issuance.

* CURRENT STATUS...A Space Weather Warning remains in effect for
  significant geomagnetic activity. Major conditions remain possible.

* FUTURE ALERTS...This is the first and only G3 Observed Space Weather
  Alert that will be issued for this event. A new Observed
  Space Weather Alert will be issued if conditions escalate to G4 or
  G5.

* IMPACTS...Power systems may experience voltage irregularities or
  false alarms on some protection devices. HF radio communication may
  be intermittent at high latitudes. GNSS users may experience increased
  uncertainty or degraded performance in disturbed ionospheric regions.
  Low-Earth-orbit satellites may experience increased drag. Aurora may
  be visible farther south than usual if local sky conditions allow.

PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should continue heightened monitoring and follow
sector-specific procedures. Power grid, aviation, satellite, GNSS, and
HF radio users should continue to monitor SWPC Warning Updates and
decision-support guidance.

$$
```

## WHY THIS PROPOSAL HELPS

One Observed Space Weather Alert at first G3 attainment records meaningful escalation without repeated threshold notifications.

- Reduces alert fatigue by suppressing duplicate G3 alerts.
- Links the observation to the continuing Warning and possible G4-G5 escalation.
- Places ongoing conditions in Warning Updates and Summaries, where context can be preserved.

**Prototype continuity — G4 Observed Space Weather Alert**

*An Observed Space Weather Alert would have been issued when the proposed 12-station Kp application first confirmed G4 conditions. Its full prototype is intentionally omitted because it would mirror the demonstrated G3 and G5 structures.*

# Observed Space Weather Alert — G5

Gannon storm reconstruction — representative prototype

## Proposed Product

```
OBSERVED SPACE WEATHER ALERT
National Weather Service Space Weather Prediction Center Boulder CO
2058 UTC Fri May 10 2024

...MAJOR GEOMAGNETIC STORM CONDITIONS OBSERVED...

* WHAT...Geomagnetic activity has reached the G5 (Extreme) level.

* OBSERVED...G5 conditions were observed at 2058 UTC during the
  2100-2400 UTC observation period.

* BASIS...This alert is based on 12/13 real-time observations from the
  official Kp magnetometer network available at the time of issuance.

* CURRENT STATUS...A Space Weather Warning remains in effect.

* FUTURE ALERTS...This is the first and only G5 Observed Space Weather
  Alert that will be issued for this event. Continued major storm
  conditions will be communicated through Warning Updates.

* IMPACTS...Power systems may experience widespread voltage-control
  issues, increased geomagnetically induced currents, and protective
  system concerns. HF radio communication may be degraded or unavailable
  over some paths. GNSS users may experience degraded accuracy, loss of
  lock, or other performance issues in disturbed ionospheric regions.
  Low-Earth-orbit satellites may experience significantly increased
  drag. Aurora may be visible much farther south than usual if local sky
  conditions allow.

PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should follow sector-specific procedures for extreme
geomagnetic activity. Power grid, aviation, satellite, GNSS, and HF
radio users should maintain heightened situational awareness and monitor
SWPC Warning Updates, Space Weather Summaries, and decision-support
guidance.

$$
```

## WHY THIS PROPOSAL HELPS

One event-level G5 Alert confirms that the storm reached the highest G-scale level without repeatedly notifying users.

- Creates a clear historical marker for peak observed severity.
- Provides operational context, expected continuation, and partner actions.
- Keeps repeated G5 periods in Warning Updates and Summaries instead of generating alert fatigue.

# Space Weather Warning Update Number 1

Gannon storm reconstruction — representative prototype

## Proposed Product

WOXX22 KWNP 110030  
WATA50

URGENT - SPACE WEATHER MESSAGE  
SPACE WEATHER WARNING UPDATE NUMBER 1  
National Weather Service Space Weather Prediction Center Boulder CO  
0030 UTC Sat May 11 2024

...SPACE WEATHER WARNING REMAINS IN EFFECT FOR MAJOR GEOMAGNETIC STORM  
CONDITIONS THROUGH 1200 UTC SUN MAY 12...

- \* WHAT...Major (G4-G5) geomagnetic storm is ongoing.
- \* WHERE...Worldwide and Earth's orbit. Middle- and higher-latitude regions are most likely to experience the greatest effects.
- \* WHEN...Warning-level geomagnetic activity is ongoing. There is a 75% chance of continued G3 or greater conditions through Saturday.  
  
Earliest Warning End: 1200 UTC Sat May 11 2024  
Latest Warning End: 1200 UTC Sun May 12 2024  
  
Shortest Remaining Duration: 12 hours  
Longest Remaining Duration: 36 hours
- \* CONFIDENCE...There is a 45% chance of additional G4 periods, with a 20% chance of additional G5 periods.
- \* IMPACTS...Power systems may experience voltage-control issues, transformer stress, and isolated equipment trips. Increased atmospheric drag may lower satellite orbits and reduce the accuracy of satellite tracking and orbit predictions. HF radio, precision satellite navigation, and some satellite communications may be intermittently degraded, particularly at middle and higher latitudes. Geomagnetically induced currents may also affect pipeline monitoring and corrosion-control systems. Aurora may be visible across much of the United States and potentially into some southern areas.

### PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should continue heightened operations, follow sector-specific procedures, and monitor SWPC updates and decision-support guidance.

- \* ADDITIONAL DETAILS...G5 conditions were first observed at 2058 UTC Fri May 10 2024, and Warning-level activity remains ongoing. This update continues the existing Space Weather Warning. Additional solar disturbances may produce renewed periods of severe to extreme geomagnetic activity into Sunday.
- \* NEXT UPDATE...No later than 1230 UTC Sat May 11 2024, or sooner if conditions change.

\$\$

WHY THIS PROPOSAL HELPS

- Warning Update Number 1 provides a stand-alone reassessment after the storm reaches G5 while retaining the familiar product structure.
- The WHEN section communicates the probability of continued warning-level activity and provides clear bounds for when the Warning may end.
  - The CONFIDENCE section distinguishes high confidence in continued significant activity through Saturday from greater uncertainty about additional disturbances into Sunday.
  - The ADDITIONAL DETAILS section documents the observed G5 escalation, confirms that the Warning remains in effect, and explains the potential for renewed activity.
  - The defined warning-end window supports staffing and mitigation decisions, while the NEXT UPDATE establishes when partners can expect another reassessment.

Ensemble guidance

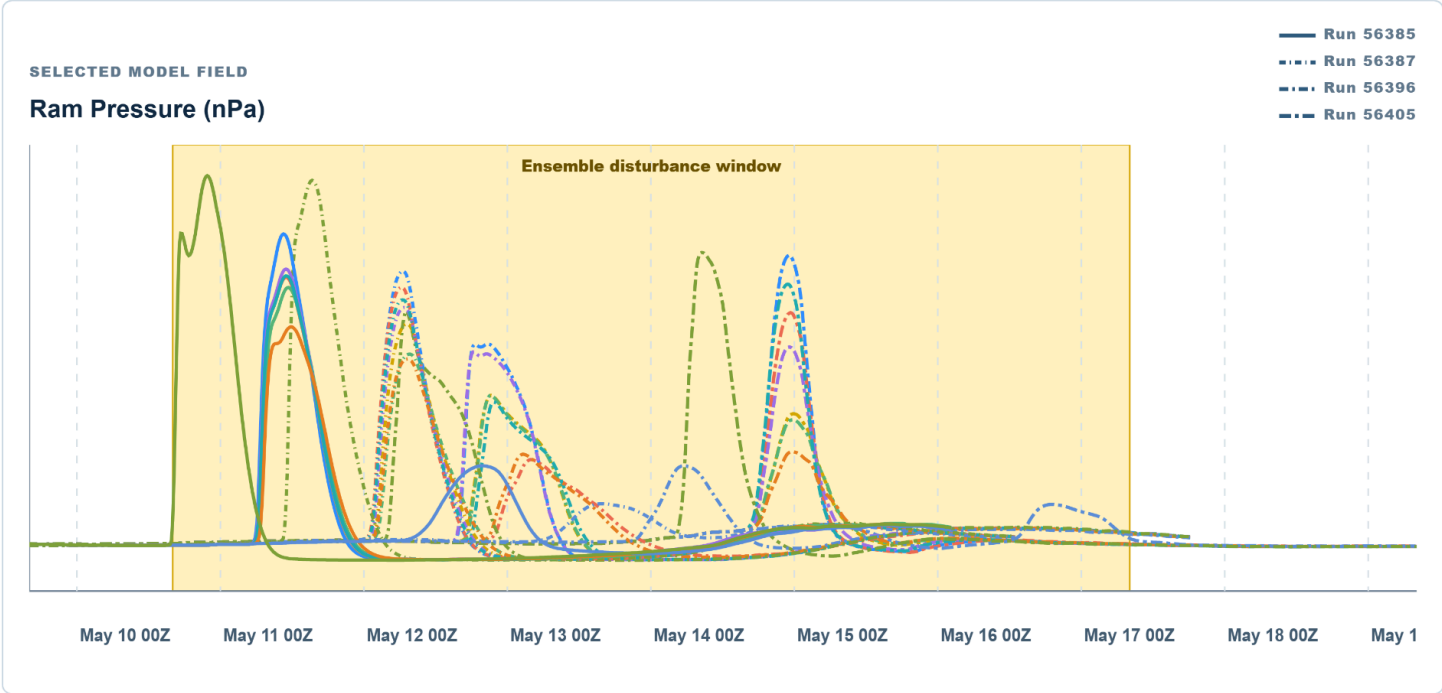
By 0030 UTC Sat May 11 2024, the product is no longer a pure forecast: G5 has been observed and warning-level activity is ongoing. Runs 56381 and 56382 have moved from forecast support into the observed event environment, so the update blends the observed escalation with remaining ensemble-supported guidance. Runs 56385, 56387, and 56396 support continued concern after the initial onset. Their fit/median arrivals center near 0623 UTC Sat May 11 2024, 0159 UTC Sun May 12 2024, and 1750 UTC Sun May 12 2024. This supports continued Warning coverage and renewed-impact concern into Sun May 12 2024. The 75% probability of continued G3 or greater conditions through 0000 UTC Sun May 12 2024 is a forecaster-assessed value based on the observed storm response plus the remaining sequence of expected impacts. It is not a direct Kp probability or a raw ensemble-member fraction.

NOAA SPACE WEATHER PREDICTION CENTER

WSA-Enlil Ensemble Guidance



|                                                                                                                    |                                                                                                   |                                                                                |                                                               |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>SCENARIO ARRIVAL WINDOW</b><br><b>1601 UTC Fri May 10 2024</b><br>through 1324 UTC Thu May 16 2024 latest onset | <b>SCENARIO MEDIAN ARRIVAL</b><br><b>1558 UTC Sun May 12 2024</b><br>Middle scenario arrival time | <b>TIMING SPREAD</b><br><b>141.4 hr</b><br>Earliest to latest scenario arrival | <b>PARAMETER RANGE</b><br><b>0.0–22.7 nPa</b><br>Ram Pressure |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------|



# Space Weather Warning Update Number 2

Gannon storm reconstruction — representative prototype

## Proposed Product

WOXX22 KWNP 111230  
WATA50

URGENT - SPACE WEATHER MESSAGE  
SPACE WEATHER WARNING UPDATE NUMBER 2  
National Weather Service Space Weather Prediction Center Boulder CO  
1230 UTC Sat May 11 2024

...SPACE WEATHER WARNING EXTENDED DUE TO ONGOING AND POSSIBLE RENEWED  
MAJOR GEOMAGNETIC STORM THROUGH 1800 UTC SUN MAY 12...

- \* WHAT...Major (G4-G5) geomagnetic storm remains possible.
- \* WHERE...Worldwide and Earth's orbit. Middle- and higher-latitude regions are most likely to experience the greatest effects.
- \* WHEN...Geomagnetic storms are ongoing. There is an 80% chance of renewed CME impact between 0600 UTC Sun May 12 and 1800 UTC Sun May 12.

Most Likely Renewed Impact: 1200 UTC Sun May 12  
Earliest Renewed Impact: 0600 UTC Sun May 12  
Latest Renewed Impact: 1800 UTC Sun May 12

Shortest Duration: 12 hours  
Longest Duration: Beyond 0000 UTC Mon May 13

- \* CONFIDENCE...There is a 60% chance of G3 conditions following renewed impact and a 20% chance of G4.
- \* IMPACTS...Power systems may experience voltage-control issues, transformer stress, and isolated equipment trips. Increased atmospheric drag may lower satellite orbits and reduce the accuracy of satellite tracking and orbit predictions. HF radio, precision satellite navigation, and some satellite communications may be intermittently degraded, particularly at middle and higher latitudes. Geomagnetically induced currents may also affect pipeline monitoring and corrosion-control systems. Aurora may be visible across much of the United States and potentially into some southern areas.

### PRECAUTIONARY/PREPAREDNESS ACTIONS...

Partners should continue heightened operations, follow sector-specific procedures, and monitor SWPC updates and decision-support guidance through 1200 UTC Sun May 12.

- \* ADDITIONAL DETAILS...Another series of coronal mass ejections may renew severe to extreme activity. This update continues the existing Space Weather Warning. If conditions decrease below warning-level concern, future products may transition to a Space Weather Statement.
- \* NEXT UPDATE...No later than 0030 UTC Sun May 12, or sooner if

conditions change.  
\$\$

WHY THIS PROPOSAL HELPS

Warning Update Number 2 preserves one continuous event narrative while shifting attention to renewed impact timing.

- Combines ongoing conditions with ensemble-supported arrival guidance.
- Provides single-number probabilities and explicit earliest, most-likely, and latest timing.
- Defines the duration risk, operational posture, and next update deadline.

Prototype continuity — additional Warning Updates

*Additional stand-alone Warning Updates would have been issued every 12 hours with the Warning extended until the call was made to ramp down*

# Space Weather Notice — coronagraph contamination

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER NOTICE  
National Weather Service Space Weather Prediction Center Boulder CO  
1400 UTC Sat May 11 2024

...ONGOING SOLAR PARTICLE STORM AFFECTING CORONAGRAPH IMAGERY...

\* WHAT...Energetic particles from the ongoing solar particle storm are degrading coronagraph imagery used to identify and analyze coronal mass ejections. SWPC is continuing to assess solar activity using available imagery, model guidance, solar wind observations, and forecaster analysis.

\* IMPACTS...Degraded imagery may limit analysis of additional coronal mass ejections and reduce confidence in their expected arrival times, duration, and potential strength. Imagery quality will be reassessed as particle levels change and additional data become available.

\* ADDITIONAL DETAILS...The active Space Weather Warning remains in effect. G3-G5 activity remains possible while coronal mass ejection-driven conditions persist. This Notice describes a data-quality limitation and does not change the active hazard assessment.

\* NEXT UPDATE...The limitation will be addressed in the next Warning Update no later than 0030 UTC Sun May 12 2024, or sooner if imagery quality changes.

\$\$

## WHY THIS PROPOSAL HELPS

A Notice about particle-contaminated imagery makes reduced forecast precision transparent during an active event.

- Distinguishes a data-quality limitation from a change in hazard severity.
- Explains how SWPC is compensating with other observations and forecaster analysis.
- Connects the limitation to an exact Warning Update deadline.

# Space Weather Summary - Mid Event

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER SUMMARY  
National Weather Service Space Weather Prediction Center Boulder CO  
1800 UTC Sat May 11 2024

...SIGNIFICANT SPACE WEATHER EVENT CONTINUES...

\* EVENT OVERVIEW...Multiple coronal mass ejections reached Earth beginning at 1639 UTC Fri May 10 2024, producing G3-G5 geomagnetic storming. Solar particle storm conditions also reached SP2. The event remains ongoing, with another series of coronal mass ejections expected to influence Earth on Sun May 12 2024.

\* WHAT HAS OCCURRED...An interplanetary shock was detected at 1639 UTC Fri May 10 2024, followed by a sudden impulse at 1645 UTC Fri May 10 2024. An Observed Space Weather Alert was issued for G5 conditions at 2058 UTC Fri May 10 2024.

Solar wind conditions became strongly enhanced, with very strong magnetic fields, sustained southward conditions, and speeds near 800 km/s. Active Region 3664 remained highly active and produced an X5.8 flare near 0123 UTC Sat May 11 2024. A halo coronal mass ejection associated with this activity remains under analysis.

\* CURRENT STATUS...Geomagnetic storming remains in progress at 1800 UTC Sat May 11 2024. Periods of G3-G4 conditions remain expected, with additional G5 periods possible if coronal mass ejection-driven conditions strengthen again.

\* WHAT IS EXPECTED...There is an 80% chance of renewed coronal mass ejection impact between 0600 UTC Sun May 12 2024 and 1800 UTC Sun May 12 2024, with the most likely impact near 1200 UTC Sun May 12 2024. There is a 60% chance that disturbed conditions will last at least 12 hours after renewed impact and a 15% chance of warning-level conditions continuing beyond 0000 UTC Mon May 13 2024.

\* ACTIVE PRODUCTS...A Space Weather Warning remains in effect through 1800 UTC Mon May 12 2024. Warning Updates, Observed Space Weather Alerts, Notices, and partner briefings will be issued as conditions and forecast expectations change.

\$\$

## WHY THIS PROPOSAL HELPS

A numbered mid-event Summary gives users one coherent account of what occurred, current conditions, and what is expected next.

- Consolidates Alerts, Warning information, flares, coronal mass ejections, and particle activity.
- Supports shift handoffs and executive briefings without requiring users to reconstruct the event.
- Carries the quantitative renewed-impact guidance into the broader event narrative.

This non-routine, event-driven Summary applies the established NWS storm-summary concept to a major space weather event. Operational adoption would require formal product-change documentation and corresponding updates to SWPC issuance guidance. [NWSI 10-502, Section 10](#) [NWSI 10-102](#)

# Space Weather Statement — G1/G2 downgrade

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER STATEMENT  
National Weather Service Space Weather Prediction Center Boulder CO  
0000 UTC Mon May 13 2024

...LINGERING GEOMAGNETIC IMPACTS POSSIBLE THROUGH 0000 UTC TUE MAY 14 2024...

\* WHAT...Lingering geomagnetic activity and elevated solar particle levels may continue as the significant space weather event subsides.

\* WHERE...Worldwide and Earth's orbit. Higher-latitude regions remain the primary area of concern.

\* WHEN...There is a 50% chance that geomagnetic activity will continue from 1800 UTC Mon May 13 2024 through 0000 UTC Tue May 14 2024.

Earliest End: 0000 UTC Tue May 14 2024  
Latest End: 1200 UTC Tue May 14 2024

Minimum Remaining Duration: 6 hours  
Maximum Remaining Duration: 18 hours

\* CONFIDENCE...There is a 40% chance of G1 conditions, with a 15% chance for G2.

\* IMPACTS...Power systems may continue to experience weak fluctuations, especially at higher latitudes. HF communication and GNSS performance may remain degraded at higher latitudes. Low-Earth-orbit satellites may continue to experience increased drag. Elevated particle levels may remain a concern for aviation, satellite, and other radiation-sensitive operations.

\* ADDITIONAL DETAILS...The Space Weather Warning has ended with the issuance of this Statement. Partners may transition from heightened operations to normal monitoring as sector-specific procedures allow.

\* NEXT UPDATE...No later than 1200 UTC Mon May 13 2024, or sooner if conditions change.

\$\$

## WHY THIS PROPOSAL HELPS

A ramp-down Statement gives users an orderly transition signal while the Warning remains active for confirmation.

- Communicates that peak G5 activity has ended while identifying residual G1-G2 risk.
- Provides explicit probabilities and timing for the expected end of lower-level conditions.
- Avoids prematurely ending the Warning before the formal stand-down product.

# Space Weather Statement — Warning-level all clear

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER STATEMENT  
National Weather Service Space Weather Prediction Center Boulder CO  
1200 UTC Mon May 13 2024

...WARNING-LEVEL SPACE WEATHER CONDITIONS HAVE ENDED...

\* WHAT...Major (G4-G5) geomagnetic storm conditions have ended. Minor geomagnetic activity and elevated solar particle levels may continue as the event subsides.

\* WHERE...Worldwide and Earth's orbit. Higher-latitude regions remain the primary area of concern for residual effects.

\* WHEN...Warning-level conditions ended by 0600 UTC Sun May 12 2024.

\* CURRENT STATUS...Solar wind conditions remain disturbed, but the geomagnetic response has weakened substantially from peak levels. Current guidance does not support renewed warning-level conditions through 1200 UTC Tue May 14 2024.

\* IMPACTS...Residual effects may continue, especially at higher latitudes and for sensitive operations. Power systems may experience weak fluctuations. HF communication and GNSS performance may remain degraded at higher latitudes. Low-Earth-orbit satellites may continue to experience increased drag. Elevated particle levels may remain a concern for aviation, satellite, and other radiation-sensitive operations.

\* ADDITIONAL DETAILS...The Space Weather Warning is no longer in effect. Partners may return to normal monitoring as sector-specific and local procedures allow. SWPC will continue monitoring the Sun and near-Earth space environment while residual activity subsides.

\$\$

## WHY THIS PROPOSAL HELPS

A warning-ending Statement gives users a clear, authoritative stand-down signal while preserving awareness of residual activity.

- Confirms that G3-G5 conditions and the Space Weather Warning have ended.
- Tells partners they may transition from heightened operations to normal monitoring.
- Bridges the period between Warning cancellation and the final event Summary.

# Space Weather Summary Number 2

Gannon storm reconstruction — representative prototype

## Proposed Product

SPACE WEATHER SUMMARY  
National Weather Service Space Weather Prediction Center Boulder CO  
1600 UTC Thu May 16 2024

...FINAL SUMMARY OF THE MAY 2024 SPACE WEATHER EVENT...

- \* EVENT OVERVIEW...The major space weather event associated with Active Region 3664 and multiple coronal mass ejection impacts from 1639 UTC Fri May 10 2024 through Sun May 12 2024 has ended. No additional significant geomagnetic impacts from this event are expected at 1600 UTC Thu May 16 2024.
- \* WHAT HAS OCCURRED...Active Region 3664 produced a series of major flares and multiple Earth-directed coronal mass ejections beginning Mon May 6 2024. As additional eruptions occurred and analysis improved, SWPC escalated its messaging from early awareness products to a formal Space Weather Warning.  
  
An Observed Space Weather Alert issued at 1647 UTC Fri May 10 2024 confirmed a sudden impulse at 1645 UTC Fri May 10 2024 following upstream shock detection at 1639 UTC Fri May 10 2024. Solar wind conditions became strongly enhanced, with very strong magnetic fields, sustained southward conditions, and speeds near 800 km/s.  
  
Geomagnetic activity subsequently reached G3, G4, and G5. G5 conditions were first observed at 2058 UTC Fri May 10 2024, with multiple G5 periods occurring into Sat May 11 2024.  
  
Solar particle storm conditions reached SP2 from Fri May 10 2024 into Sat May 11 2024, with elevated particle levels continuing into the recovery period. Active Region 3664 remained flare-productive as it approached the west limb, producing additional X-class and M-class flares.
- \* CURRENT STATUS...Solar wind conditions have weakened substantially from the peak of the storm, and geomagnetic activity has returned to routine monitoring levels. Active Region 3664 has rotated beyond the west limb, reducing the likelihood of additional Earth-directed impacts from this region.
- \* ADDITIONAL DETAILS...All Watches, Warnings, and event-related Space Weather Statements associated with this event are no longer in effect. Partners may return to normal monitoring of SWPC products and sector-specific guidance. SWPC will issue new products if additional space weather activity develops.

\$\$

## WHY THIS PROPOSAL HELPS

A final Space Weather Summary provides one authoritative record of the event and confirms the return to normal operations.

- Consolidates the event evolution, peak severity, timing, and major operational concerns.
- Confirms that all event-related Watches, Warnings, and Statements have ended.
- Provides formal closure for partners, shift handoffs, verification, and future review.